

AIDC MARKET REPORT

Power & Storage Sales Strategy – the 40-company competitive map

Prepared 10 August 2026 · **Corpus** 40 Profiler dossiers, researched 7–10 August 2026 · **Method** No new research – every citation is a source already catalogued in a dossier
Verification 572 extracted claims · 353 report claims adversarially re-checked · 20 findings corrected pre-publication · **Classification** Internal – sales strategy

PRESENTATION STYLE – Intelligence Community Briefing. Key judgments carry explicit confidence levels; reported fact and labelled assessment never mix.

HOW TO READ THIS REPORT

Facts cite their dossier source as (*Source label, date – Company dossier*). Items marked **ANALYSIS** come from dossier *strategy read* sections – labeled inference, carrying (High)/(Moderate)/(Low) confidence tags where the dossier holds them – and are never blended with sourced fact. Figures restate quantities already cited in the body text; each carries its own source line. Field notes are not sources and none are cited here.

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1. KEY JUDGMENTS

12	9	3	≈\$735–760B
Key judgments, each confidence-tagged	Rated High confidence	Rated Moderate confidence	Big-four 2026 capex underwriting the pipeline

1. **(High – report synthesis; the underlying Equinix dossier read is tagged Moderate)**

Power – not demand, capital, or chips – is the binding constraint on the AI buildout: ≈\$130B of blocked or delayed US projects in Q1 2026, 0.3% Northern Virginia vacancy, and buyers formally denominating demand in gigawatts, corroborated across §2's sourced order books. Every account in this pipeline is paying for *time-to-power*; every product should be sold as schedule compression. (§2)

2. **(High)** The demand base is enormous and partially pre-sold – Microsoft's \$678B RPO grows faster than its capex – but its financing is circular by construction (NVIDIA↔OpenAI↔Oracle↔Amazon). Sell into the urgency while tracking the fragility through order books, which would show a pause first. (§2, §8.4)

3. **(High)** Turbine and transformer scarcity is now sold as a product: GE Vernova's deposit-backed slots price 10–20 points above backlog against 2031 delivery, and reservation fees are back at Siemens Energy for the first time since the early 2000s. It is scarcity rent, not moat – relief lands 2027–30 – but transformers stay tight into the 2030s (≈30% US deficit, prices +77% since 2019). The queue prices every alternative a storage or power seller offers. (§3)

4. **(High)** Behind-the-meter generation-plus-storage is the *proven* architecture for gigawatt AI campuses: xAI ran ≈1.0 GW almost entirely off self-built gas buffered by the world's largest Megapack fleet, and Crusoe productized the same model with the deepest published energy stack. Storage is embedded in every BTM variant – the campus bill of materials is the account map. (§4)

5. **(High)** FERC's Talen precedent killed uncompensated behind-the-meter nuclear co-location and pushed hyperscaler nuclear front-of-meter at premium prices (≈\$112/MWh restart, ≈\$70/MWh existing) – and nearly all contracted nuclear delivers 2027–2030s. The gas-plus-storage bridge owns this decade. (§4)

6. **(High)** 800 VDC is a vendor-list reset on a published clock (first shipments 3Q26, Kyber racks 2027): Delta is structurally ahead, ABB is winning the medium-voltage layer, no AC incumbent has a shipping 800 VDC switchboard, and the architecture makes rack-level storage (BBU and supercapacitor, ≈\$2.8B → ≈\$7.1B by 2033) non-discretionary content. Qualification is happening now, inside NVIDIA reference designs and digital twins – not in RFPs. (§5)

7. **(High)** The BESS league table reset in 2025–26: Sungrow took both Wood Mackenzie crowns, Chinese integrators hold ≈76% of a >100 GW market that flipped from glut to tightness, and storage became the profit center for Sungrow and Tesla Energy – and the growth engine of CATL's ESS line. (§6)

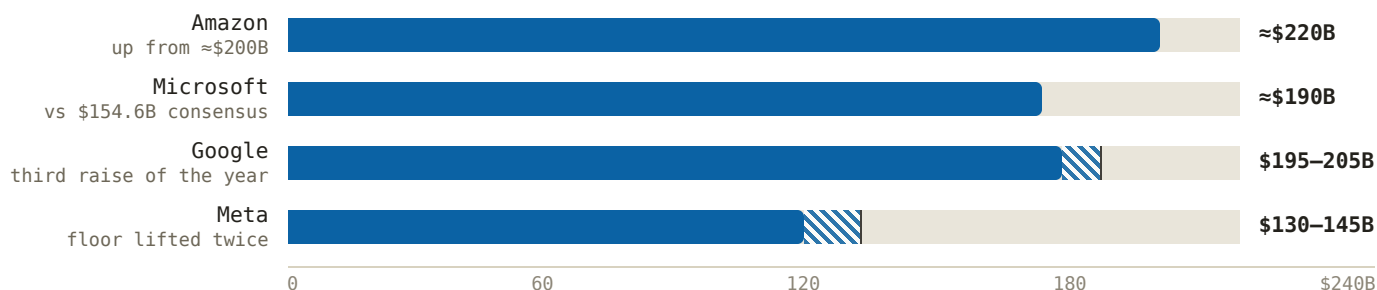
8. **(High)** The FEOC, tariff and 1260H wall is the compliant lane's pricing umbrella: LGES, Samsung SDI, Panasonic, Fluence and Tesla's onshoring convert regulation into share, while Chinese leaders route around it via licensing (CATL–Ford), offshore cells and minority JVs. Domestic content, UL fire certification and cybersecurity attestation are now qualifying criteria. (§6)

- 3. **(High)** AI-data-center storage demand arrives through four distinct doors – utility-scale via developer counterparties, campus MV-BESS via energy-first developers and colos, UPS-replacement via electrical contractors, and rack BBU/supercapacitor via the NVIDIA ODM chain. The hyperscalers split: Google is the most storage-forward (owns Intersect, anchored 30 GWh iron-air), Microsoft structurally buys grid instead of batteries, Amazon and Meta buy through counterparties. Qualifying *which door* an account opens is the first sales question. (§6, §8.1)
- 4. **(High)** Craft labor is the scarcest input in the US buildout – four years to make a craftsman, ≈\$13.5M/MW of craft-led spend – and prefabrication is the industry's answer. Labor-per-MWh is now a spec: factory-integrated blocks (Megablock, Fibrebond eHouses, Modular Power Solutions rooms) are winning on installed-MW lead time, not component price. (§7)
- 5. **(Moderate)** Displacement windows are open in 2026: Wärtsilä's storage counterparty changes hands ≈Q3 2026, Powin's orphaned 25+ GWh fleet is a services land-grab in progress, the AC incumbents' 800 VDC gap is provable, and Hitachi's own capacity relief does not land until 2027–28. The gap years are the selling window. (§8.2)
- 6. **(Moderate)** The risk stack to monitor on a calendar, not a hunch: circular financing, ≈20% near-term conversion of the 307 GW US pipeline, ratepayer and political backlash (PJM auctions at the cap; Meta's Louisiana docket), the xAI permitting-litigation template, and scarcity-rent normalization as 2027–28 capacity lands. The armed post-earnings Routines are the monitoring instrument. (§8.3–8.4)

2. THE DEMAND BACKDROP – WHY THIS PIPELINE EXISTS

The AIDC buildout is no longer sized in dollars alone. The buyers themselves now denominate demand in gigawatts, and every theme in this report flows from that.

Figure 1: 2026 AI capital expenditure, the big four (\$ billion, company guidance)



Source: Amazon Q2 2026 earnings, 2026-07-30; CNBC Q3 FY2026, 2026-04-29; Alphabet Q2 2026 8-K, 2026-07-22; CNBC Q2 2026 coverage, 2026-07-29 – Amazon, Microsoft, Google and Meta dossiers. Note: hatched extension marks a guided range; solid marks a point estimate. Combined big-four 2026 capex ≈\$735–760B.

Table 1: The 2026 capex stack – the signal behind each number

BUYER	2026 CAPEX	SIGNAL	CITATION
Amazon	~\$220B	Largest single-company AI program; AWS backlog \$244B, up 40%	Amazon Q2 2026 earnings, 2026-07-30; FY2025 8-K, 2026-02-05 – Amazon dossier
Microsoft	~\$190B	Commercial RPO \$678B, up 84%; Azure crossed \$100B/yr	CNBC Q3 FY2026, 2026-04-29; FY2026 Q4 results, 2026-07-29 – Microsoft dossier
Google	\$195–205B	First \$400B revenue year; quarterly FCF –\$5.9B; 2027 guided to increase significantly	Alphabet Q2 2026 8-K, 2026-07-22; Q4/FY2025 8-K, 2026-02-04 – Google dossier
Meta	\$130–145B	Hyperion at 5 GW design / \$50B+; Q2 capex ate ≈98% of operating cash flow	CNBC Q2 2026 coverage, 2026-07-29; Richland Parish expansion, 2026-07-13 – Meta dossier

DEMAND IS NOW STATED IN GIGAWATTS, ON THE RECORD

- **OpenAI:** ≈\$1.4T / ≈30 GW of acknowledged commitments. The 10 GW-by-2029 target was surpassed in commitments by April 2026, with more than 3 GW added in one trailing 90-day window; planned compute spend through 2030 was raised to \$750B (*Reuters factbox*, 2025-10; *Stargate scaling update*, 2026-04; *WSJ via Yahoo*, 2026-07-22 – OpenAI dossier). Vendor by vendor: NVIDIA ≥10 GW, Broadcom 10 GW, AMD 6 GW, AWS ≈2 GW Trainium, Cerebras 750 MW (*Tunguz commitment tally*, 2025-10 – OpenAI dossier; the AWS ≈2 GW line also per Amazon's Q1 2026 8-K, 2026-04-29).
- **Amazon** put gigawatts in an earnings filing: Anthropic up to 5 GW of Trainium, OpenAI ≈2 GW, 2.1M+ AI chips deployed in 12 months (*Q1 2026 8-K exhibit*, 2026-04-29 – Amazon dossier).
- **NVIDIA** – the demand engine – booked FY2026 revenue of \$215.9B, up 65%, with data center at \$193.7B, and nearly doubled its own supply commitments quarter on quarter from \$50.3B to \$95.2B (*Q4/FY2026 results*, 2026-02-25 – NVIDIA dossier). **ANALYSIS – HIGH** Third-party projections of ≈233

- GW of cumulative new AI data-center power demand 2025–2030 rest heavily on NVIDIA's shipment cadence.
- **The neocloud tier is the marginal buyer:** CoreWeave crossed 1 GW active across 49 data centers with 3.5+ GW contracted and a \$99.4B backlog, up 284% (*Q1 2026 results, 2026-05-07 – CoreWeave dossier*); xAI went from 0 to 1.0 GW of nameplate compute draw in three years (*SpaceX Form S-1, 2026-05-20 – xAI dossier*).

HOW REAL IS IT? PRE-SOLD VERSUS SPECULATIVE

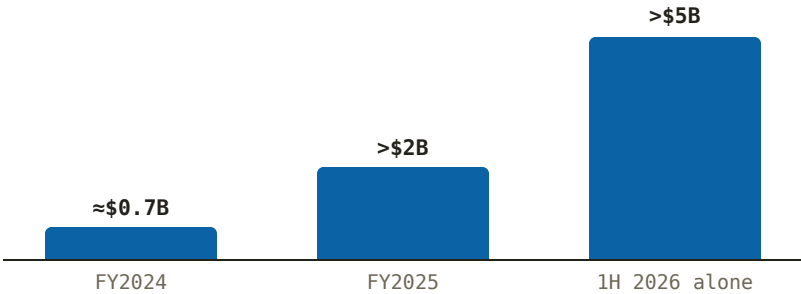
- **ANALYSIS – HIGH** Microsoft's demand case is the best-documented of the big four: a \$678B backlog growing faster (+84%) than capex means the buildout is substantially pre-sold – the OpenAI \$250B Azure commitment alone underwrites years of Fairwater capacity (*Microsoft dossier*).
- **ANALYSIS – MODERATE** Oracle's \$638B RPO overstates firm demand: roughly half is press-attributed to a single ≈\$300B OpenAI contract starting 2027, \$75B is GPU-prepayment structures, and ≈0.3 GW is operational versus more than 9 GW projected by 2029 (*Oracle dossier*). Oracle's FY2026 free cash flow was –\$23.7B, funded by \$43B of debt plus \$5B equity with ≈\$40B more planned (*Q4 FY2026 earnings, 2026-06-10 – Oracle dossier*).
- **ANALYSIS – HIGH** The financing model is circular by construction: NVIDIA invests up to \$100B as OpenAI deploys NVIDIA systems; Amazon invests \$50B while selling \$138B of cloud; Oracle borrows (≈\$96B reported) against OpenAI receivables; Microsoft's \$4.1B quarterly equity-method losses imply ≈\$11.5B quarterly OpenAI losses (*OpenAI dossier*).
- The debt market keeps repricing the risk downward anyway: CoreWeave's GPU-backed coupon went from ≈15% in 2023 to investment-grade SOFR+225bps (DDTL 4.0, \$8.5B, 2026) (*globaldatacenterhub, 2026-03-15 – CoreWeave dossier*).

POWER IS THE GATE – THE FACT THIS WHOLE REPORT MONETIZES

≈\$130B	0.3%	>70 MW	≈20%
US projects blocked or delayed on power, Q1 2026 – a record	Northern Virginia data-center vacancy	Amsterdam connection moratorium threshold; Singapore rations releases	Of the 307 GW US pipeline converting near-term – the bulk is 2028+

Source: Equinix dossier (vacancy, moratoria, blocked-project total – **ANALYSIS – MODERATE**); Eaton dossier (pipeline conversion – **ANALYSIS – MODERATE**).

Figure 2: The order books confirm it – GE Vernova data-center Electrification orders (\$ billion)



Source: GE Vernova Q2 2026 press release, 2026-07-22 – GE Vernova dossier. Note: the 1H 2026 column is a half-year figure against two full years – the ramp is steeper than the bars show.

Table 2: Corroborating order books across the supplier set

SUPPLIER	ORDER-BOOK SIGNAL	GROWTH	CITATION
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SUPPLIER	ORDER-BOOK SIGNAL	GROWTH	CITATION
ABB	First-ever \$7B+ Electrification quarter; triple-digit DC growth	+60%	<i>Q2 2026 release, 2026-07-16 – ABB dossier</i>
Vertiv	Backlog \$15.0B on ≈2.9x book-to-bill	+109%	<i>FY2025 8-K, 2026-02-11 – Vertiv dossier</i>
Siemens Energy	Q3 FY26 Gas Services orders €10.0B (73 turbines / 15 GW); record €162B group backlog	2.65x b-t-b	<i>Q3 FY26 earnings release, 2026-08-05 – Siemens Energy dossier</i>
Schneider	Data-center demand grew triple-digit	triple-digit	<i>H1 2026 release, 2026-07-30 – Schneider dossier</i>
Hitachi Energy	Q1 FY2026 orders; backlog above ¥10T (≈\$68B)	+87% YoY	<i>Investing.com, 2026-07-29 – Hitachi Energy dossier</i>

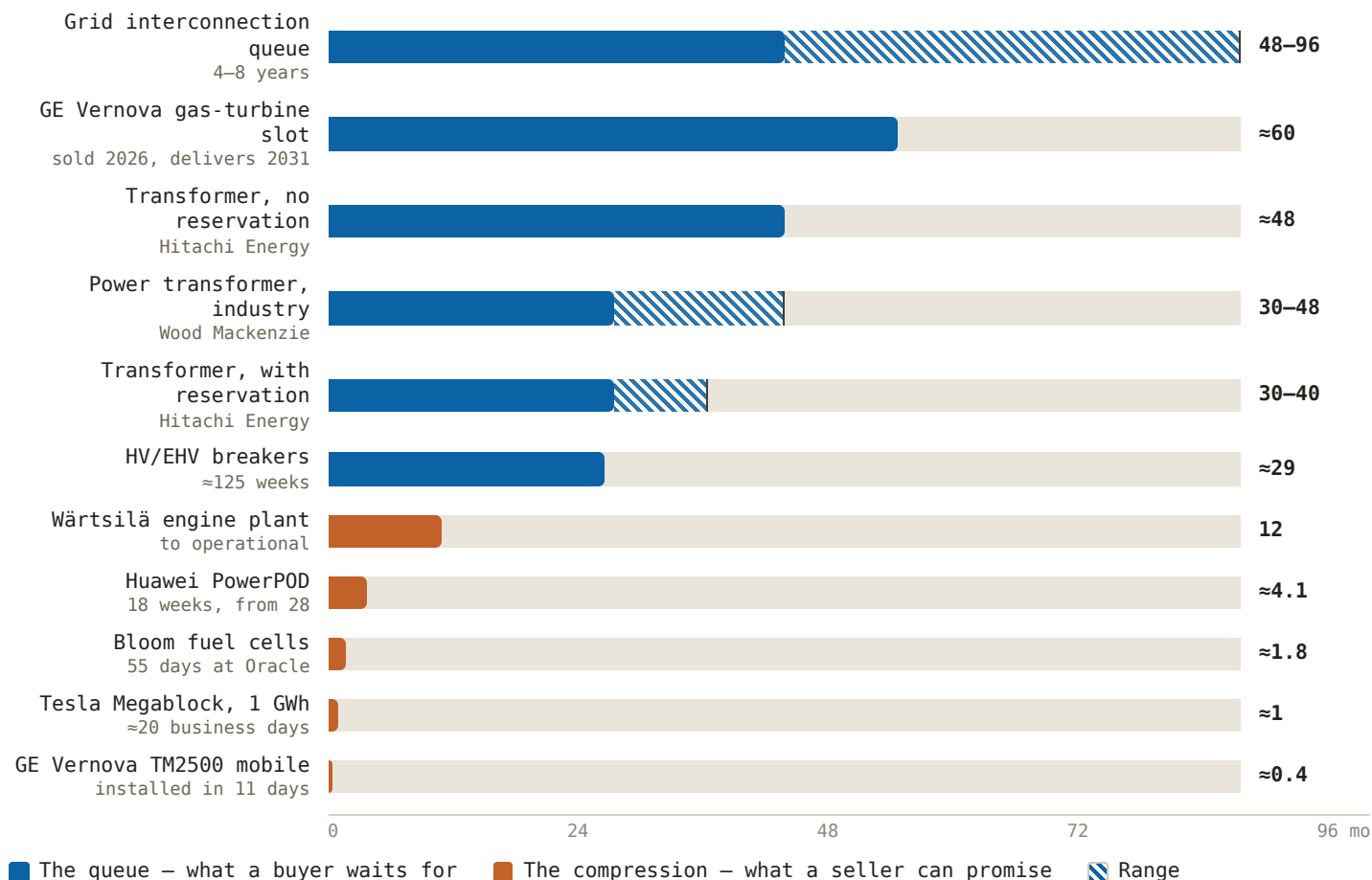
SALES TRANSLATION

Every buyer in Table 1 is paying for *time-to-power*, not electrons. The pipeline's job is to position power and storage products as schedule compression – that framing recurs in every section below.

3. TURBINE & TRANSFORMER SCARCITY ECONOMICS

Scarcity in prime movers and grid equipment is the single most quantified phenomenon in the dossier set – and it is being **monetized as a product** (delivery slots), not just endured.

Figure 3: The master clock – queue times versus what compression products promise (months)



Source: Hitachi Energy dossier (Nikkei Asia, 2024-05; Wood Mackenzie, 2024-07 / 2025-02); GE Vernova dossier (Turbomachinery Magazine, 2026-07; gas-power product pages); Quanta dossier (Q2 2026 release, 2026-07-30); Wärtsilä, Huawei, Bloom, Tesla dossiers as cited in the body text. Note: one shared axis in months – the sub-2-month bars are drawn to true scale, which is the point. Bloom's own comparator is interconnect waits of up to ~7 years.

TURBINES: THE SLOT IS THE PRODUCT

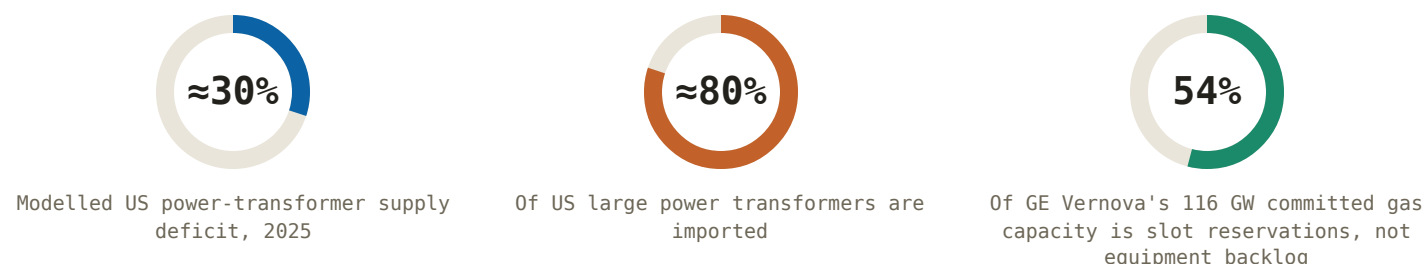
- GE Vernova holds **116 GW of committed gas capacity** – 53 GW of equipment backlog plus 63 GW of *deposit-backed slot reservations* now selling against **2031 delivery** (Turbomachinery Magazine, 2026-07 – GE Vernova dossier). Reservations price 10–20 points above the existing backlog, taken with ~20% cash down that converts to orders – the clearest scarcity-rent evidence in the sector (December 2025 Investor Update transcript, 2025-12-09). Output ramps ~20 GW/yr → 24 GW (2028) → actions toward 30 GW (2030), and GE Vernova expects to be “largely sold out of 2030 deliveries by end of 2026” (Utility Dive, 2025-12).
- **Siemens Energy:** large-frame slots nearly full through 2028, and slot-reservation fees are back for the first time since the early-2000s gas boom (POWER Magazine, 2025-02 – Siemens Energy

dossier). Unit sales nearly doubled in one year – 194 turbines in FY25 against 100 in FY24 (*Siemens Energy release, 2025-11-14*).

- The demand shock is documented from the buyer side too: xAI signed ≈\$3.7B of turbine commitments in four months of 2026 – \$805M plus \$925M purchase agreements through 2029 plus a ≈\$2.0B mobile-fleet acquisition (*SpaceX Form S-1, 2026-05-20 – xAI dossier*). **ANALYSIS** That makes it one of the largest mobile-turbine buyers anywhere, in a market where every AIDC developer is chasing the same units. Utilities are in the same queue: Xcel ordered 10 × SGT6-5000F (2,088 MW) for Texas – “dispatchable power is no longer optional” (*Siemens Energy release, 2025-10-01*).
- **ANALYSIS – HIGH** GE Vernova's pricing power is real but it is *scarcity rent, not durable moat*: reservations imply ≈\$3,000/kW combined-cycle pricing, roughly 3x in three years, while Siemens Energy doubled unit sales and Mitsubishi calls its own +30% capacity add “not enough” – competitor supply lands in the same 2028–2030 windows GE Vernova is now selling (*GE Vernova dossier*).

TRANSFORMERS: THE TIGHTER, LONGER BOTTLENECK – AND THE MORE DEFENSIBLE FRANCHISE

Figure 4: Transformer scarcity, in three ratios



Source: Wood Mackenzie, 2025-02 (deficit) and Siemens Energy \$1B US investment release, 2026-02-03 (import share) – Hitachi Energy and Siemens Energy dossiers; Turbomachinery Magazine, 2026-07 – GE Vernova dossier (63 GW of 116 GW). Note: each ring is a share of a stated whole; growth rates are shown as tiles below, not rings.

+77%

US transformer prices since 2019; constraints modelled “well into the 2030s”

10–20 pts

Premium GE Vernova charges on reserved slots over existing backlog

\$57.9B

Hitachi Energy backlog – ≈2.9x annual revenue, on FY2025 orders of \$32.8B

13.4%

Hitachi Energy adjusted EBITA margin, up from 6.1% in FY21

- Hitachi Energy – the world's largest transformer maker – quotes 30–40 month waits, up to four years without a reservation, rationing access through reservation agreements (*Nikkei Asia, 2024-05 – Hitachi Energy dossier*). The E.ON framework – up to \$700M, 20,000+ transformers under a capacity-reservation model – is first-party proof that transformer capacity is now allocated years ahead by reservation (*Hitachi Energy release, 2025-07-28*).
- Its financials show the scarcity monetization directly: FY2025 orders \$32.8B (a 28% CAGR from \$12.4B in FY21), backlog \$57.9B, adjusted EBITA margin 13.4% against 6.1% in FY21 (*Hitachi Ltd FY2025 annual results presentation, 2026-04-27*); Q1 FY2026 orders +87% year on year, backlog above ¥10T (≈\$68B) (*Investing.com, 2026-07-29*).
- GE Vernova paid \$5.275B to consolidate Prolec GE – five US transformer plants, ≈10,000 employees – and is now the most North America-weighted top-tier transformer maker (*Prolec completion, 2026-02-02*). **ANALYSIS – HIGH** The transformer bottleneck is the more defensible franchise versus turbines: 2.5–4-year lead times modelled to persist toward 2030, and GE Vernova's data-center electrification orders quintupled in 18 months.

- Breakers are the third choke point: ≈125-week lead times are throttling load-center schedules – Quanta's Hyosung HICO JV (HV/EHV breakers to 800 kV, Canonsburg PA) is aimed squarely at them (*Quanta Q2 2026 release, 2026-07-30 – Quanta dossier*).

Table 3: The capacity response is real but back-loaded – and that is the selling window

SUPPLIER	COMMITTED SPEND	LANDS	SCOPE
Hitachi Energy	\$9B+ global / \$1B+ US	2028	The \$457M South Boston VA plant, billed as the largest US transformer factory (<i>release, 2025-09-04</i>)
Siemens Energy	+30–50% capacity	FY26–28	Transformer and turbine capacity expansion (<i>CMD 2025, 2025-11-20</i>)
Quanta Services	\$500–700M	2028	Nearly doubles HV transformer capacity (<i>2026 Investor Day, 2026-03-31</i>)
Schneider Electric	>\$700M	through 2027	Eight US sites (<i>release, 2025-03-25</i>)
ABB	\$200M	2026+	European medium-voltage capacity (<i>release, 2026-05-11</i>)

ANALYSIS – MODERATE Cyclical risk is real but back-loaded: ≈\$1.8B of announced US transformer investment across the industry all lands 2027–28, and pricing power (+60–80% transformer inflation since 2020) normalizes if utilities' ordering behavior does (*Hitachi Energy dossier*). **ANALYSIS** Siemens Energy's "measured expansion plus shareholder returns" choice supports pricing power but prolongs the bottleneck (*Siemens Energy dossier*).

SALES TRANSLATION

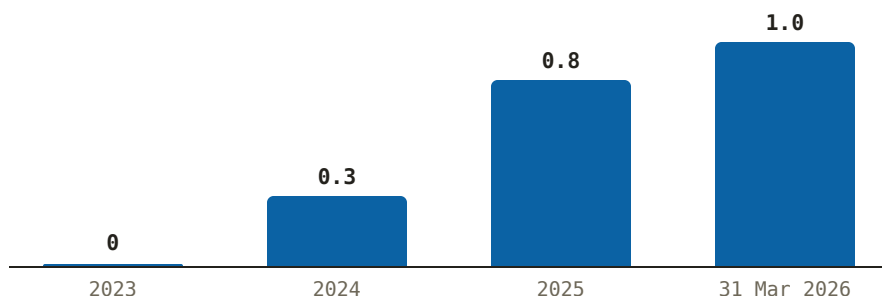
Scarcity is the master clock of this market, and it prices every alternative. Bloom's entire AEP positioning is that fuel cells deploy in 6–9 months against ≈3 years for turbines **ANALYSIS – HIGH** – the same arithmetic sells BESS bridge power, prefab substations, and anything else that converts queue time into revenue time. Prospects without reservation frameworks are queued four years at Hitachi's window; every one of them is a lead for whoever can compress that schedule. The flip side: quotes priced off 2026 scarcity rents need re-validation for 2028+ delivery, when the announced capacity wave lands.

4. BEHIND-THE-METER POWER – THE XAI / CRUSOE TEMPLATE

Behind-the-meter (BTM) generation is the industry's answer to the interconnection queue, and the dossier set contains both templates: **xAI**, the extreme vertically self-built case, and **Crusoe**, the productized merchant-developer case. Everything else in this section is the market forming around what those two proved.

4.1 THE XAI TEMPLATE – SELF-BUILD AT ANY COST

Figure 5: Colossus nameplate compute draw – 0 to 1.0 GW in three years (GW)



Source: SpaceX Form S-1, 2026-05-20 – xAI dossier. Note: powered “almost entirely” by self-deployed behind-the-meter generation; the filing confirms the campus “is capable of operating entirely” off self-built generation. The grid is the supplement, not the source.

- **The hardware:** ≈460 MW of documented on-site gas across the two campuses – 12 SMT-130s at Memphis, 7 Titan 350s at Southaven (*SemiAnalysis*, 2025-12); a permitted permanent 1.2 GW / 41-turbine plant under construction, with 69 temporary mobile turbines exiting August 2026 – July 2027 under a Mississippi DEQ order (*CNBC*, 2026-03-10); more than 240 Megapacks at Colossus 1 alone – the S-1 claims the world's largest Megapack deployment and the first gigawatt-scale Megapack installation, enabling full off-grid operation during grid stress (*S-1*, 2026-05-20).
- **The procurement velocity:** ≈\$3.7B of turbine commitments in four months of 2026, plus Tesla Megapack purchases of \$191M (2024), \$506M (2025) and \$34M (Q1 2026) – ≈\$731M through Q1 2026 – plus another ≈\$269–295M by SpaceX in Q2 2026, for ≈\$731M–1B cumulative, figures varying by source (*S-1*, 2026-05-20; xAI dossier).
- **ANALYSIS – HIGH** Speed-to-power is the differentiator, not model quality: BTM gas plus Megapack buffering bypasses the queue entirely; 64–91-day cluster bring-ups are the fastest independently corroborated; TVA and MLGW grid allocations (≈300 MW) cover only a fraction of load (xAI dossier).
- **The risk half of the template, equally documented:** the NAACP/SELC federal Clean Air Act suit, four different turbine counts with different evidentiary statuses (35 documented, 15 permitted backup-only, 27 alleged, 41 permitted), and a **\$399M self-recorded litigation accrual** (*CNBC*, 2026-04-14). **ANALYSIS – MODERATE** Related-party procurement is structural: vendors negotiating with SpaceXAI should assume **Tesla gets first look at any storage scope** (xAI dossier).

4.2 THE CRUSOE TEMPLATE – ENERGY-FIRST DEVELOPMENT AS A PRODUCT

- Crusoe's procurement list reads like this report's supplier index: 29 × GE Vernova LM2500XPRESS (≈1 GW, SCR, 5-minute start) (*Crusoe release*, 2025-01); ≈750 MW of Bergen reciprocating engines (438 MW firm plus 310 MW LOI) with Piller dynamic stabilization for GPU transients (2026-06-03); **5 GW of ON.energy medium-voltage “AI UPS” BESS** (2026-07-21); 12 GWh of Form Energy iron-air 100-

hour storage from 2027 (2026-03-24); and two nuclear plays – Aalo's 10 MWe INL pilot in 2027 scaling to 50 MWe plants by end-2029 (2026-07-30), and Blue Energy's gas-bridge-2028-to-nuclear-2031 path (Crusoe dossier).

- **The proof case:** Stargate Abilene phase 1 live on OCI within one year of construction start (2025-09-30); the 900 MW Microsoft Campus 2 pairs two 336 MW-IT buildings with a 900 MW on-site plant **plus medium-voltage battery storage** (2026-03-27). **ANALYSIS – HIGH** Time-to-power execution at scale is the genuine edge: 200+ MW energized within a year, a substation in under six months.
- **The caveats:** ≈4.9 GW contracted against a claimed 40+ GW pipeline (2026-07); single-customer decisions reshaped the company twice in 2026 (the Oracle/OpenAI expansion halt; Google's Wyoming ejection); **ANALYSIS – MODERATE** the gas fleet runs on minor “permit by rule” instruments with a pending major permit that would rank among Texas's largest fossil plants.

4.3 THE MARKET THAT FORMED AROUND THE TEMPLATES

Table 4: Behind-the-meter posture by buyer – the same problem, five different answers

BUYER	ARCHITECTURE	STORAGE CONTENT	CITATION
Oracle	A per-site power menu: Project Jupiter (Doña Ana NM) up to 2.45 GW of Bloom fuel cells, <i>displacing planned gas turbines and diesel gensets</i> ; Shackelford County fully off-grid on reciprocating gensets (115 MW live, 2.0 GW site); Abilene on ERCOT plus ≈360 MW of on-site GE Vernova turbines	Site-by-site; “we pay our own way on energy” policy – on-site generation or Oracle-funded upgrades, no ratepayer pass-through	<i>Project Jupiter announcement, 2026-04-27; Oracle post, 2026-01-26 – Oracle and Bloom dossiers</i>
OpenAI	Stargate sites span the whole BTM menu: Shackelford 2.0 GW gas microgrid; Doña Ana 2.2 GW gas plus Bloom; Port Washington WI 1.3 GW at ≈70% solar/wind/battery; Saline Township MI 1.4 GW on DTE grid plus batteries	Battery-heavy at two of four named sites; LGES Vertech supplies the DTE program	<i>Epoch AI Stargate tracking – OpenAI dossier</i>
Meta	Prometheus (≈1 GW, online 2026) on 200–400 MW of BTM gas plus AEP transmission; the Hyperion/Entergy package adds seven gas plants (≈2.3 GW approved, ≈5.2 GW proposed)	Three grid-scale batteries in the Entergy package; storage bought utility-side, not by Meta	<i>Data Center Frontier; Meta, 2026-07-13 – Meta dossier</i>

BUYER	ARCHITECTURE	STORAGE CONTENT	CITATION
Google	Bought the model outright – the ≈\$4.75B Intersect Power acquisition brings ≈2.2 GW solar and 2.4 GWh BESS in-house for “energy parks”: data centers co-located with BTM solar-plus-storage to bypass queues. First park at Haskell County TX	640 MW solar + 1.3 GWh BESS at Haskell County; 1 GW of data-center demand response across five utilities	<i>PV Tech, 2026-03; Google, 2026-03-19 – Google dossier</i>
Equinix	Proves the colo tier buys firm power too: the industry's largest colo fuel-cell fleet (Bloom, past 100 MW across 19+ sites), ≈1.25 GW of advanced-nuclear commitments and LOIs including Oklo 500 MW – the first colo-operator SMR deal	A 20-year take-or-pay with Central Georgia EMC where Equinix funds 100% of grid upgrades	<i>Equinix dossier, 2025-08-14 / 2026-08-06</i>
Huawei Red Sea	The off-grid ceiling: 400 MW PV plus 1.3 GWh BESS, 100% renewable, 21+ months at 99.9% critical-load reliability	1.3 GWh – the entire firming layer is storage	<i>ESS News, 2024-09-18 – Huawei Digital Power dossier</i>

ANALYSIS – MODERATE The queue is the strategic enemy Google is engineering around – the playbook rivals will copy (Google dossier). **ANALYSIS – HIGH** Equinix's choices function as procurement standards for the whole colo tier (Equinix dossier).

4.4 THE PRODUCTIZED SUPPLY RESPONSE – WHAT IS ACTUALLY FOR SALE

Table 5: Onsite-power products, compared on the claim that sells them

OFFERING	WHAT IT IS	THE HEADLINE CLAIM	CITATION
Siemens Energy × Eaton 500 MW block	Standardized SGT-800-based grid-independent campus power – redundant turbines plus battery storage, no diesel	99.99%+ availability; up to two years off time-to-power. ANALYSIS – HIGH the most productized onsite-power offering from any major OEM	<i>Siemens Energy release, 2025-06-03 – Siemens Energy dossier</i>
GE Vernova aeroderivatives	LM6000, LM2500XPRESS and TM2500 mobile units	“Two weeks to power generation”: 5-minute starts, >99.8% availability, TM2500 installed in 11 days; Chevron “power foundries” target up to 4 GW for data centers by 2027	<i>GEV gas-power pages; Chevron partnership, 2025-06-15 – GE Vernova dossier</i>

OFFERING	WHAT IT IS	THE HEADLINE CLAIM	CITATION
Wärtsilä engines	Reciprocating-engine power plants	789 MW of first US data-center projects booked in 2025; a 429 MW / 24-engine utility order (Jan 2026); large plants operational in 12 months; 20–35% lower fuel consumption than turbines (company claim); Pure DC Ireland as Europe's first off-grid DC microgrid	<i>Wärtsilä FY2025 bulletin and product pages – Wärtsilä dossier</i>
Bloom Energy fuel cells	Solid-oxide Energy Servers, now shipped “800V DC ready”	55-day Oracle deployment beat its own 90-day commitment; AEP firmed 1 GW at ≈\$2.65B – ANALYSIS the best public pricing benchmark for fuel-cell prime power at ≈\$2.65/W. First \$1B+ revenue quarter in Q2 2026; the Brookfield partnership expanded fivefold to \$25B	<i>Bloom dossier, 2026-01-15 / 2026-04-13; 8-K, 2026-07-28; release, 2026-06-30</i>
ABB × VoltaGrid	62 flywheel synchronous condensers for BTM AI campuses, delivered as prefab eHouse packages	ANALYSIS – HIGH The strongest independent validation that AI load volatility is purchasable – stabilization is now its own product category	<i>Engineering.com, 2026-03-25 – ABB dossier</i>
FlexGen	Campus BESS with grid-forming PCS	Millisecond response to AI swings, deployable in under a year against 4–8-year queues	<i>FlexGen solutions page – FlexGen dossier</i>

ANALYSIS – HIGH Bloom's core sale is time: 55–90 days against interconnect waits of up to ≈7 years – but the headline pipeline overstates firm demand (the Oracle 2.8 GW figure is a ceiling with ≈1.2 GW contracted, and management disclaims more than ≈6 months of visibility).

4.5 THE FRONT-OF-METER COUNTERWEIGHT – NUCLEAR AND THE FERC LINE

Figure 6: The nuclear price ladder – what a restart costs versus existing output (\$/MWh, analyst estimates)



Source: Jefferies via Energy Connects, 2024-09; Utility Dive, 2025-06 – Constellation dossier. Note: analyst estimates, not disclosed contract prices. **ANALYSIS – HIGH** Constellation's pricing power – a \$20–50/MWh premium band, 920 MW of PPAs in a single quarter – is the market-defining asset.

- FERC's November 2024 rejection of the Talen/Amazon BTM nuclear ISA, upheld April 2025, set the national precedent: **behind-the-meter co-location without cost allocation is dead**. Amazon's restructured front-of-meter PPA, ramping to 1,920 MW of Susquehanna through 2042, is the workaround template (*Utility Dive*, 2024-11; Talen release, 2025-06-11 – Amazon dossier). **ANALYSIS – MODERATE** That pushes incremental firming demand toward grid-side assets – including storage. FERC later ordered PJM to write transparent co-location and large-load tariff rules (*Utility Dive*, 2025-12-18 – Constellation dossier).
- Meta's contracted nuclear now totals ≈7.7 GW across TerraPower, Oklo, Vistra and Clinton – the largest corporate nuclear commitment (*Meta*, 2026-01-09 / 2025-06-03); Amazon targets 5+ GW by 2039; Microsoft holds TMI (2027) plus the world's first fusion PPA (Helion, ≥50 MW targeting 2028) (*Microsoft dossier*). But nearly all of it delivers **2027–2030s** – nothing nuclear firms AI load this decade at scale.

SALES TRANSLATION

The BTM bill of materials is the account map. Every campus needs generation (turbines, engines, fuel cells), stabilization (flywheels, Piller-class systems, grid-forming BESS), a medium-voltage storage buffer sized to GPU transients, and increasingly long-duration storage as interconnection currency. Storage is embedded in every variant: xAI's Megapack fleet, Crusoe's 5 GW ON.energy MV-BESS plus 12 GWh iron-air, Microsoft's Campus-2 MV storage, Google's energy parks, Stargate's battery-heavy sites, Meta's Entergy batteries. The gas-plus-storage bridge owns 2026–2030 while nuclear matures; the objections to pre-handle are the xAI permitting record and Texas permit-by-rule scrutiny, both now shaping every BTM procurement conversation.

5. THE 800 VDC TRANSITION

NVIDIA is not a power vendor, but it writes the spec everyone else must build to – and in May 2025 it rewrote the data-center power architecture around 800 VDC. This is the sector's vendor-list reset, and reset moments are when share moves.

5.1 THE ARCHITECTURE AND ITS CLOCK

+157%	–45%	+5 pts	–30%
More power through the same copper versus 415 VAC	Copper reduction – a legacy 1 MW rack carries up to 200 kg of busbar	End-to-end efficiency, up to	Total cost of ownership to

Source: NVIDIA 800 VDC whitepaper, 2025-10-13 – NVIDIA dossier. Note: vendor-published figures; each measures a different quantity and they are not additive.

The partner slate is the qualification list. Unveiled at Computex on 2025-05-20 with Delta, LITEON, Megmeet, Eaton, Schneider, Vertiv, Infineon and TI; expanded at OCP on 2025-10-13 to the **grid-equipment tier** – ABB, GE Vernova, Hitachi Energy, Siemens and Heron Power (NVIDIA 800VDC blogs, 2025-05-20 / 2025-10-13 – NVIDIA dossier).

Figure 7: The published clock – the qualification window is open now and closes with Kyber

2025	2026	2027	2028
2025-05	Computex – 800 VDC architecture unveiled; ODM partner slate named. Qualification window opens.		
2025-10	OCP – slate expanded to the grid-equipment tier; two-tier storage hierarchy published (rack supercapacitors, facility BESS)		
3Q 2026	First 800V HVDC shipments – NVIDIA (Vera Rubin) and Google adopt first (TrendForce, 2026-06-15)		
Q3 2026	Delta ±400 VDC enters mass production		
Q4 2026	Delta 800 VDC small-volume production		
Q1 2027	LITEON 800V mass production – ASIC customers first; GPU clients “potentially Q2 2027”		
2H 2027	Kyber racks – 576 Rubin Ultra GPUs, ≈600 kW/rack; full-scale 800 VDC production coincides. Window closes.		
2027	GE Vernova's hyperscaler decision on 1,000 solid-state transformers, if spec is met		
Feynman era	“1 MW IT racks and beyond”		

Source: TrendForce, 2026-06-15; Tom's Hardware GTC record, 2025-03-18; NVIDIA 800 VDC whitepaper; GE Vernova December 2025 investor update – NVIDIA, Delta, LITEON and GE Vernova dossiers.

THE ARCHITECTURE MANDATES STORAGE

NVIDIA prescribes a two-tier energy-storage hierarchy – supercapacitors at the rack for millisecond transients, facility-level BESS at the interconnection for second-to-minute smoothing (NVIDIA whitepaper, 2025-10-13) – and GB300 NVL72 already productizes rack-level smoothing (≈65 J/GPU in power shelves, up to –30% peak grid demand) (NVIDIA blog, 2025-10). Power-quality and storage content is becoming **non-discretionary rack content**.

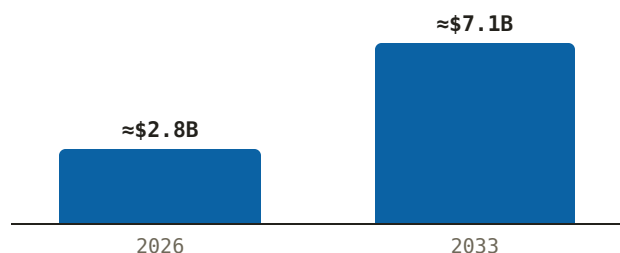
5.2 THE ODM TIER – WHO IS ACTUALLY AHEAD

Table 6: The three named power ODMs, compared

VENDOR	POSITION TODAY	800 VDC STACK	VOLUME TIMING	THE CATCH
Delta	#1 in NVIDIA's AI-server power ecosystem; first-qualified across GB200/GB300 shelf sets; ≈70% of Blackwell-platform PSUs; reported >50–60% of AI-server AC/DC	The deepest published: 660 kW in-row power rack embedding 480 kW of BBU per rack ; SST converting MV AC to 800 VDC at 98.5%; SiC e-fuse with <3 μs cutoff; claimed >4% grid-to-chip gain to ≈92%	±400 VDC MP Q3 2026; 800 VDC small volume Q4 2026; true volume 2027	AI data-center power and cooling crossed 50% of revenue in Q2 2026 – concentration cuts both ways. ANALYSIS – HIGH structurally advantaged into the 800V generation
LITEON	#2 at ≈30% AI-server power share against Delta's 60%+	The BBU franchise is its structural edge: BBU shipments ran ≈30x in 2025 and moved from optional (GB200) to standard (GB300/Rubin); its 800V rack embeds BBU plus supercapacitor shelves	800V MP Q1 2027 – ASIC customers first, GPU clients “potentially Q2 2027”	A step behind Delta on the clock. The \$919M McKinney TX plant moves non-China capacity to 60% by end-2026. ANALYSIS – HIGH its strongest products become non-discretionary content
Megmeet	The only mainland-China power supplier NVIDIA named for GB200 NVL72	Kyber 800V sidecar first-mover (claims +5% efficiency, –45% copper, –30% TCO); full grid-to-GPU chain shown including a >98.5% SST	Claimed first-mover; no disclosed volume date	By its own statements AI orders were immaterial through 2025 ; FY2025 profit fell 66.6% funding the pivot; three limit-downs in four July sessions. ANALYSIS – HIGH the 2026-08-27 H1 report is the first hard test

Source: WAWT, 2026-04-20; KGI, 2025-07-01; CommonWealth; TrendForce, 2026-06-15; Delta Q2 results, 2026-07-29; LITEON release, 2026-07-13; SmBom, 2025-05-22; Megmeet releases 2025-05-21 / 2026-04-13 – Delta, LITEON, Megmeet and NVIDIA dossiers.

The BBU cell layer beneath them. Samsung SDI's ≈KRW 2T BBU chain runs via Simplo, with both Samsung SDI and Panasonic reported **supply-short in BBU cells** (*The Elec*, 2026-07 – *Samsung SDI dossier*); Panasonic's BBU plus supercapacitor-CBU strategy matches the rack-side architecture (*Panasonic*, 2026-03-25); EVE showed a full-matrix 800V BBU portfolio at OCP China 2026 – **ANALYSIS – MODERATE** real differentiation potential but no named wins, so 2027+ optionality (*EVE dossier*).

Figure 8: The rack-storage market the architecture creates – AI-server BBU (\$ billion)

Source: TrendForce – NVIDIA dossier. Note: each GB300 cabinet uses ≈5 BBU modules plus more than 300 supercapacitors, and supercapacitor supply is a genuine bottleneck. **ANALYSIS – MODERATE** NVIDIA's rack storage legitimizes and expands the market faster than it cannibalizes it. Independent analysis (SemiAnalysis) separately projects ≈\$11B of power-rack and ≈\$13B of solid-state-transformer markets forming alongside.

5.3 THE INCUMBENTS' DILEMMA – AND THE GRID-TIER LAND GRAB

ANALYSIS – MODERATE The transition is a *net-content* question for the AC incumbents: independent analysis projects 800 VDC **contracts the centralized UPS market** – a core Vertiv, Eaton and Schneider franchise – while creating the power-rack and SST markets above. **No incumbent has yet named a shipping 800 VDC switchboard product** (Vertiv and Eaton dossiers).

Table 7: Where each incumbent actually stands on 800 VDC

VENDOR	SHIPPING OR ANNOUNCED	COMMERCIAL DATE	READ
ABB	HiPerGuard, the industry's first static MV UPS (2.5–25 MW, 98%, now a 34.5 kV class that removes a conversion stage); SACE Infinitus, the world's first IEC-certified solid-state circuit breaker	Shipping	ANALYSIS – HIGH Winning the medium-voltage layer; the architecture incumbent for the next rack generation. Eaton, Schneider and Vertiv compete at low voltage
Vertiv	Full 800 VDC portfolio – centralized rectifiers, DC busway, rack DC-DC, DC-compatible backup; PowerDirect Rack already in shelf/BBU territory	2H 2026	Ahead of Kyber, but a portfolio release is not a switchboard. Binding risk is execution
Eaton	OCP reference architecture staking supercapacitor fast-cycle backup plus ORv3 busbar, packaged as Beam Rubin DSX; market-first detection of GPU-induced subsynchronous oscillations	Reference stage	Strong on the transient problem; the switchboard gap stands
Schneider	800 VDC sidecar – racks to 1.2 MW with integrated energy storage	No commercial date	The design layer is the real edge: NVIDIA co-developed designs pull hardware into projects <i>before</i> competitive bidding ANALYSIS – HIGH
GE Vernova	Three AI-factory 800 VDC reference designs (grid-connected, islanded, bridging; 13.8 kV AC → 800 VDC via SSTs); R&D cost-share with a hyperscaler committed to buy 1,000 SSTs from 2027 if spec is met	SST 12–24 months out	ANALYSIS – MODERATE If the spec lands, GE Vernova leapfrogs the AC-switchboard incumbents; if it slips, the SST market forms around Heron Power-class challengers

VENDOR	SHIPPING OR ANNOUNCED	COMMERCIAL DATE	READ
Hitachi Energy	Co-designed grid-to-rack conversion for 800 VDC racks from 3 MW to GW-class	Co-design stage	ANALYSIS – MODERATE Its edge is owning the grid side of the interface; its gap is no UPS or rack installed base

Source: ABB dossier, 2026-07-16 / 2025-10-13; Vertiv release, 2025-10-13 and 2025-03-18; Eaton, 2025-10-13 / 2026-03-16 / 2025-09-09; Schneider, 2025-10-13; GE Vernova whitepaper, 2025-10-24 and December 2025 investor update; Hitachi dossier, 2025-10-13.

SPECIFICATION NOW HAPPENS IN THE DIGITAL TWIN

ABB's SimReady assets, Eaton's Beam Rubin DSX, Schneider's ETAP/AVEVA models and GE Vernova's reference designs all live inside NVIDIA's Omniverse DSX Blueprint. Spec wins happen in the twin **before procurement opens** (ABB, 2026-06; Eaton, 2026-03-16; Schneider, 2026-03-16).

5.4 THE LONG TAIL – SMALLER ENTRANTS RIDING THE RESET

- **Sinexcel** stood up a dedicated AIDC division in June 2025, developing HVDC systems and SSTs on its APF/SVG power-electronics heritage. Current shipping exposure is a 36 kW HVDC rectifier module – third-party reporting says supplied **to Vertiv**, giving indirect NVIDIA exposure – with claimed ≈100 W/in³ density against an ≈70 industry average; HVDC and SST launches expected 2026. **ANALYSIS – MODERATE** Pre-revenue optionality; the H1 2026 report, due **2026-08-11**, is the checkpoint for first AIDC revenue disclosure (海之理财, 2025-10-09 – Sinexcel dossier).
- **Bloom** now ships every Energy Server “800V DC ready” for direct DC integration (datasheet, 2026-02); **Jinko's** Sunny 365 AIDC edition claims 800V HVDC compatibility for solar-plus-storage supply (release, 2026-08-04).

SALES TRANSLATION

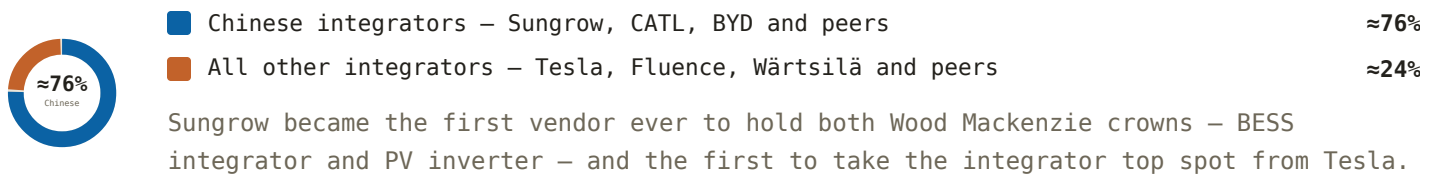
800 VDC is a **vendor-list reset on a published clock** – qualification windows are open now through 2027, and the spec is being won inside NVIDIA reference designs and the Omniverse twin, not in RFPs. For a power or storage seller: (1) BBU, supercapacitor and rack-storage content is becoming mandatory – the storage conversation now starts *inside* the rack; (2) any product not in a named ecosystem tier needs a route in through someone who is – the Sinexcel-to-Vertiv module path is the template; (3) incumbents defending centralized-UPS revenue are the displacement targets, and their own dossier reads concede the switchboard gap; (4) near-term calendar: Sinexcel H1 on 08-11 and Megmeet H1 on 08-27 are both direct reads on whether the challenger tier is converting the reset into revenue.

6. BESS COMPETITIVE DYNAMICS

The storage market crossed 100 GW of annual installs in 2025 (*ESS News, 2026-07-14 – Fluence, Tesla and Wärtsilä dossiers*) and flipped from glut to tightness, with the ESS cell market up 94.6% to 612 GWh (*InfoLink via ESS News, 2026-02-09 – Hithium and EVE dossiers*). Five dynamics matter for the pipeline: the league-table reset, the hardware arms race, the geopolitical fault line, the integrator shakeout, and the convergence of BESS onto the data center itself.

6.1 THE LEAGUE TABLE, POST-RESET

Figure 9: Chinese integrators took roughly three-quarters of a market above 100 GW



Source: Wood Mackenzie inaugural Global BESS Integrator Comprehensive Ranking, July 2026, via *ESS News, 2026-07-14 – Sungrow, Tesla, Fluence and Wärtsilä dossiers*. Note: rankings are scope-dependent – Benchmark's system-shipment cut has BYD #1 for 2025 (>60 GWh, 13%) over Tesla (46.7 GWh). **ANALYSIS – MODERATE** Treat single-ranking claims with caution and always name the ranking body.

Table 8: The 2026 integrator league table and what sits behind each position

RANK	INTEGRATOR	SHARE SIGNAL	THE LOAD-BEARING FACT
1	Sungrow	Both crowns	ESS overtook inverters as its largest segment – 41.8% of revenue at 36.5% gross margin. ANALYSIS – HIGH storage is now the profit center
2	Tesla	46.7 GWh (2025)	Tesla Energy became Tesla's profit engine – deployments +49%, Q4 margin 28.7%
3	CATL	27.1% of cells	ESS line hit RMB 53.26B in H1 2026, up 87.5%, ≈20% of group revenue
4	BYD	>60 GWh (2025)	#1 on Benchmark's system-shipment cut at 13% share – the ranking-scope caveat in one line
7	Fluence	≈4%	Record \$6.4B backlog including ≈\$850M of data-center orders and master supply agreements with two hyperscalers
top-10	Wärtsilä	EUR 719M book	The Western “pure-play” tier – now folding into a 50/50 JV with RCT Solutions

Source: *ESS News, 2026-07-14 (Wood Mackenzie ranking); Benchmark via BYD dossier; Sungrow, Tesla, CATL, Fluence and Wärtsilä dossiers; Teslarati, 2026-01; Energy-Storage.News, 2026-07; Fluence Q3 FY2026, 2026-08-05.*

Figure 10: The cell layer is consolidating more slowly – ESS cell share, H1 2026

Source: CnEVPost, 2026-08-04 (H1 2026 CATL); CATL dossier (share trajectory, Hithium and EVE at roughly 12% each); InfoLink, 2026-02-09 – EVE dossier (FY2025 cell ranking: Hithium Top-2, EVE #3, displaced from #2 by Hithium). Note: Hithium and EVE shares are approximate and every wedge is direct-labeled.

+94.6%

ESS cell market growth to 612 GWh – glut flipped to tightness

41.8%

Sungrow revenue from ESS, at 36.5% gross margin – now its largest segment

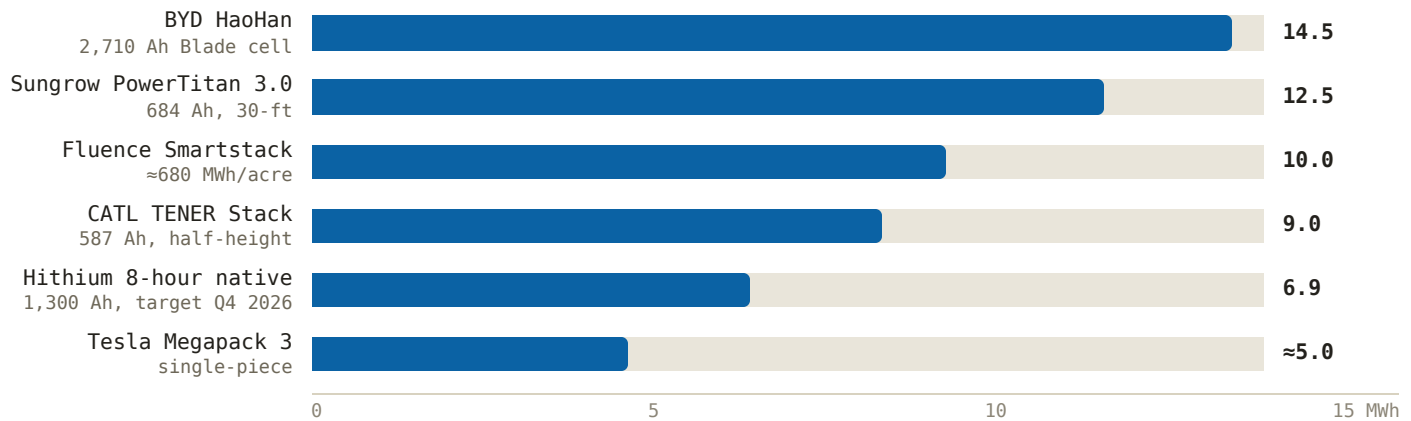
+87.5%

CATL ESS revenue growth, H1 2026, to RMB 53.26B

+95–110%

EVE H1 2026 profit guidance as pricing power returned

6.2 THE HARDWARE ARMS RACE – BLOCKS, CELLS, SAFETY

Figure 11: The block-size leapfrog – energy per factory-integrated block (MWh)

Source: pv magazine, 2025-09-19 (BYD), 2025-06-10 (Sungrow), 2025-05-08 (CATL), 2026-06-09 (Hithium); Fluence, 2026-06-23; ESS News, 2025-09-09 (Tesla) – BYD, Sungrow, CATL, Hithium, Fluence and Tesla dossiers. Note: block energy is one axis of the race; footprint, cell capacity and installed labor are the others – see Table 9.

Table 9: The cell and block race, on the specifications each vendor leads with

VENDOR	FLAGSHIP BLOCK	CELL	THE CLAIM THAT SELLS IT
BYD	HaoHan, 14.5 MWh	2,710 Ah Blade	233 kWh/m ³ ; lifecycle cost claim below CNY 0.1/kWh
Sungrow	PowerTitan 3.0, 12.5 MWh	684 Ah	First mass-producible 684 Ah cell; SiC PCS; 12.5 MWh in a 30-ft container. DNV-witnessed 20 MWh burn test with no propagation, suppression disabled
Fluence	Smartstack, 10 MWh	–	≈680 MWh/acre; up to 40% lower balance-of-plant
CATL	TENER Stack, 9 MWh	587 Ah	“5-year zero degradation”; road-legal half-height containers. TENER Sodium is the first field-validated sodium-ion BESS – 30+ MWh configurations, 15,000 cycles, 30-year warranty

VENDOR	FLAGSHIP BLOCK	CELL	THE CLAIM THAT SELLS IT
Hithium	6.9 MWh, 8-hour native	1,175 → 1,300 Ah	≈Cell 1175Ah was the first mass-produced kAh-class cell; the 1,300 Ah system targets Q4 2026
EVE	Grid station, Feb 2026	628 Ah Mr.Big	First mass-produced 600 Ah+ prismatic; world's first 628 Ah grid station; more than 30 GWh of 2027 orders in hand; claims ~50% installation workload versus 314 Ah systems
Tesla	Megapack 3, ≈5 MWh	LFP (Sparks / LGES)	Brookshire 50 GWh/yr line live 2026-08-06 – the fastest factory bring-up in company history. Megablock bundles four Megapack 3s with factory-integrated MV transformer and switchgear
Samsung SDI	SBB 2.0	LFP prismatic	World-first pass of UL's Indoor Large-Scale Fire Test for UPS batteries – framed explicitly as a hyperscaler procurement requirement

Source: as Figure 11, plus CATL releases 2026-06/07 and *Energy-Storage.News*, 2024-11 (sodium-ion); PR Newswire, 2025-06 and *pv magazine*, 2026-06-09 (Hithium); EVE releases 2024-12 / 2026-02-10 / 2026-07; Electrek, 2026-08-06 (Brookshire); PR Newswire, 2024-11 (Sungrow burn test); Samsung SDI, 2026-07-14. **ANALYSIS – MODERATE** Sodium-ion is a post-lithium transition competitors must now answer (CATL dossier). Safety is now a procurement gate, not a datasheet line.

6.3 WHERE THE MEGA-ORDERS ARE – AND WHAT AI ADDS

The Gulf giga-orders set the scale bar. Masdar's 5.2 GW / 19 GWh round-the-clock project runs on CATL TENER – 19 GWh, the largest single BESS order (*ESS News*, 2025-01-20) – with BYD adding 11.275 GWh for EWEC and Sungrow winning 7.5 GWh plus 2.6 GW of inverters on the same program (*BYD*, 2026-07-09; *ESS News*, 2026-05-22). BYD also took a 12.5 GWh SEC award, and Sungrow's 7.8 GWh Alghaz plant – the world's largest grid-forming BESS – was won next door to Huawei's Red Sea showcase (*pv magazine*, 2025-02-17 / 2024-07-16; *ESS News counter-datapoint*, 2025-12-18 – Huawei dossier).

AI data centers are now a quantified storage demand channel – not pipeline talk:

- **1.6 GWh** of Samsung SDI's H1 2026 ESS shipments went to AI data centers (*SNE via CnEVPost*, 2026-08-04).
- **Fluence** carries ≈\$850M of data-center orders plus master supply agreements with **two hyperscalers** inside its record \$6.4B backlog – **ANALYSIS – HIGH** a contracted channel (*Fluence dossier*; Q3 FY2026, 2026-08-05).
- **LGES Vertech's** 6 GWh / 1.5 GW DTE program exists to serve the Oracle-managed OpenAI campus at Saline Township (*LGES*, 2026-05-27).
- **Google** anchored the largest battery project by energy capacity announced anywhere – 300 MW / **30 GWh** of Form Energy iron-air via Xcel – plus the largest US solar-plus-storage PPA (Steel River: full build 2.5 GW solar and 2.9 GWh BESS) (*Google*, 2026-02-24; *pv magazine*, 2026-07-15). **ANALYSIS – HIGH** Google has chosen ownership over procurement – the hyperscaler most likely to shape, not just consume, the storage market.

Table 10: Hyperscaler storage posture – who to sell to, and through whom

BUYER	POSTURE	EVIDENCE	WHO ACTUALLY HOLDS THE OEM DECISION
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BUYER	POSTURE	EVIDENCE	WHO ACTUALLY HOLDS THE OEM DECISION
Google	Owner	Bought Intersect Power (≈\$4.75B, 2.2 GW solar + 2.4 GWh BESS); anchored 30 GWh Form Energy iron-air; largest US solar-plus-storage PPA	Google itself – the only hyperscaler that has internalized development
Amazon	Counterparty	Utility-scale OEM decisions sit with developer counterparties – AES at Bellefield, Primergy	ANALYSIS – HIGH The developer , not the hyperscaler – the most established hyperscaler-linked storage channel
Meta	Counterparty	Buys utility-side via tolled projects (Enbridge Cowboy: 200 MW / 1,600 MWh on Tesla batteries); reserved up to 1 GW / 100 GWh of Noon long-duration storage	ANALYSIS – MODERATE Meta's utility counterparties – Entergy, Enbridge – not Meta
Microsoft	Structural non-buyer	Runs Fairwater Atlanta with <i>no on-site generators or UPS</i> – 4x9 availability at 3x9 cost. No major 2025–26 standalone BESS procurement has surfaced	ANALYSIS – HIGH Has structurally chosen grid procurement over on-site storage. The Dublin grid-interactive UPS pilot with Eaton is the door still ajar

Source: PV Tech, 2026-03 and Google releases 2026-02-24 / 2026-03-19; Amazon dossier; Utility Dive, 2026-05-21 and Meta, 2026-04-27; Microsoft dossier, 2025-11-12.

6.4 THE GEOPOLITICAL FAULT LINE – THE FEOC AND TARIFF UMBRELLA

The wall is rising in layers. US tariffs on Chinese battery imports step from 7% to **26% in 2026** (*ESS News*, 2025-06-06 – *Hithium dossier*). CATL sits on the DoD 1260H list with Pentagon procurement bars phasing in from June 2026 and a full bar in October 2027, pushing its US strategy to the License-Royalty-Service route – Ford's \$3.5B BlueOval plant is the template (*CnEVPost*, 2026-07-24; *electrive*, 2026-06-18 – *CATL dossier*). EVE was reportedly added to 1260H in June 2026 while the LG/Tulip ITC action threatens cylindrical imports – **ANALYSIS – HIGH** the US market is closing on EVE from three directions, and minority stakes, licensing and offshore cells are the workaround template. A draft FCC rule would outright ban Chinese energy inverters, and the EU is phasing Huawei and Sungrow inverters out of funded projects by April 2027 (*Reuters via US News*, 2026-06-30; *Euronews*, 2026-05-04 – *Huawei dossier*).

Table 11: The compliant lane – who is converting regulation into share, and how fast

VENDOR	THE ONSHORE MOVE	MILESTONE DATE	READ
LGES	140 GWh ESS backlog; the \$4.3B Tesla Megapack 3 LFP cell award at Lansing; more than KRW 3T of new ESS contracts including AI-data-center deals	Lansing cells from 2027	H1 2026 ESS shipments +357% , ≈86% North America. ANALYSIS – HIGH converting EV overcapacity into an ESS franchise faster than any peer; ANALYSIS – MODERATE ex-AMPC operating losses make credit-dependence the vulnerability

VENDOR	THE ONSHOREING MOVE	MILESTONE DATE	READ
Samsung SDI	Two billion-dollar-class US ESS deals in four months – KRW 1.5T prismatic, more than KRW 2T LFP as SBB 2.0	US LFP cells from Oct 2026	Demand exceeding capacity from 2028. ANALYSIS – MODERATE StarPlus conversion is the pivotal execution item
Panasonic	¥800B FY2029 data-center storage target, described as “a minimum commitment”; Kansas EV lines physically repurposed to data-center cells	FY2029 target	Claims ≈80% share in data-center distributed power – self-reported and unverified
Fluence	First US domestic-content shipments; FEOC compliance expected on OBBBA deadlines	Shipping since Sep 2025	ANALYSIS – MODERATE Constraint is execution, not demand – ≈\$400M of Q3 revenue delayed by new US manufacturing
Tesla	First Megapacks with US-made LFP cells at Sparks, built with CATL equipment	Shipping 2026	ANALYSIS – MODERATE Onshoring reduces but does not eliminate CATL dependence
Hithium	Mesquite TX assembly (10 GWh/yr) plus a €400M Spanish gigafactory	First US shipments Aug 2025	Chinese-owned but building inside the wall – the hedge both ways

Source: LGES dossier and release 2026-05-27; Samsung SDI dossier; Panasonic, 2026-03-25 / 2026-06-08; ESS News, 2025-09-05 and Energy-Storage.News, 2025-11 (Fluence); TeslaNorth, 2026-07-31; Energy-Storage.News, 2025-08 and ESS News, 2026-03-11 (Hithium).

6.5 THE INTEGRATOR SHAKEOUT – AND BESS CONVERGING ON THE DATA CENTER

The Western pure-play tier is consolidating. Powin's Chapter 11 in June 2025 vaporized 25+ GWh of EVE's contracted US demand (ESS News, 2025-06-11); FlexGen bought Powin's hardware and software IP for \$36M and **ANALYSIS – HIGH** deliberately exited the hardware lane for software and services (HybridOS: 65+ configurations, 22 vendors) (FlexGen dossier); Wärtsilä is folding its storage business (EUR 719M order book) into a 50/50 JV with RCT Solutions after a 60% order collapse – **ANALYSIS – HIGH** a managed retreat, and **ANALYSIS – MODERATE** the counterparty transition around the ≈Q3 2026 closing is a live displacement opening for rivals.

Table 12: BESS is walking into the data center through four doors

DOOR	THE PROOF CASE	WHO YOU SELL TO	SIZE SIGNAL
UPS replacement	Rosendin + FlexGen BESSUPS – MV utility-scale battery UPS outside the data hall, patents combined	Electrical contractors	Displaces the centralized-UPS franchise 800 VDC is already contracting
Campus MV-BESS	Crusoe's 5 GW ON.energy “AI UPS” commitment, ERCOT ride-through validated	Energy-first developers and colos	5 GW from a single developer commitment

DOOR	THE PROOF CASE	WHO YOU SELL TO	SIZE SIGNAL
Reference-design embedding	Fluence Smartstack selected for Siemens' NVIDIA AI-data-center reference architecture – load smoothing, black start, ride-through. Vertiv's EnergyCore Grid (1 MW–200+ MW) crosses from DC-power incumbency into the integrator lane ANALYSIS – MODERATE	The reference-architecture owner, before procurement opens	Wins the spec, not the bid
Inside the rack	Each GB300 cabinet uses ≈5 BBU modules plus more than 300 supercapacitors; supercapacitor supply is a genuine bottleneck	The NVIDIA ODM chain – Delta, LITEON, Megmeet	≈\$2.8B (2026) → ≈\$7.1B (2033) – Figure 8

Source: Business Wire, 2025-06-03 (BESSUPS); Crusoe, 2026-07-21; Energy-Storage.News, 2026-06-02 (Smartstack in the Siemens reference architecture); Vertiv dossier, 2025-12-03; TrendForce – NVIDIA dossier.

SALES TRANSLATION

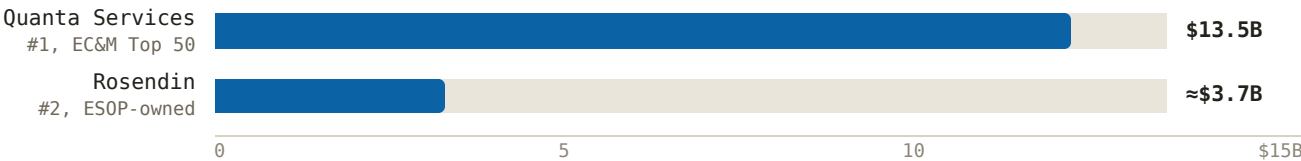
Know which of the four doors an account opens – utility-scale (sold to developers and IPPs, not hyperscalers; Amazon proves it), campus MV-BESS (sold to developers like Crusoe and to colos), UPS-replacement (sold with electrical contractors – the Rosendin/FlexGen pattern), or rack-level BBU and supercapacitor (sold through the NVIDIA ODM chain). The FEOC and tariff umbrella is the compliant lane's pricing power – lead with domestic content, UL fire certification and grid-forming references. Displacement windows open now: Wärtsilä's JV transition, Powin's orphaned fleet (25+ GWh of supported base up for services), and every incumbent whose 800 VDC story is still a slide.

7. THE CRAFT-LABOR BOTTLENECK

The scarcest input in the US buildout is not silicon or steel – it is trained craft labor on a four-year development cycle. The dossiers quantify both the constraint and the industry's answer, prefabrication, and both are sales-relevant.

4 years To make a craftsman – the CEO's rule at the #1 US electrical contractor	≈\$13.5M Craft-led spend per MW of data center, excluding outside-the-fence power	220–240% Guided 2026 growth in Quanta's Technology & Load Centers end market	10–20 yrs How far out customers now give Rosendin work forecasts
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Figure 12: The US electrical-contractor tier is a duopoly of scale – electrical revenue (\$ billion)



Source: EC&M 2025 Top 50, 2025-06 – Quanta dossier. Note: Quanta is roughly 3.7x the #2 on electrical revenue. Rosendin's total company revenue is ≈\$5.6B across 12,000 employees.

THE CONSTRAINT, QUANTIFIED

- **Quanta Services** is the #1 US electrical contractor at \$13.5B of electrical revenue. Its platform: 69,500 employees including 50,000+ craft, more than 80% self-perform, +15,500 hires in a trailing year, a 2,300-acre lineman college, and ≈\$250M a year on training (*2026 Investor Day, 2026-03-31*). Management prices craft-led spend at **≈\$13.5M per MW** of data center, excluding outside-the-fence power, claiming 80–90% balance-of-plant coverage. Q2 2026 revenue ran \$9.56B, up 41%, on a record \$53.4B backlog – **ANALYSIS – HIGH** which is *neither floor nor firm*: the NiSource ≈3 GW program is mostly excluded and data-center MSA work runs book-and-burn outside backlog, while RPO is only \$33.6B (*Investor Day plus Q2 2026 call, 2026-07-30 – Quanta dossier*).
- **Rosendin** (#2, ESOP, ≈\$5.6B revenue across 12,000 employees) reports customers providing work forecasts **10–20 years out** (*Construction Dive, 2026-01-28*), with 4M+ electrician man-hours of annual labor capacity (*Rosendin sector page*). **ANALYSIS – HIGH** Rosendin is near-pure-play exposure to the AI electrical buildout – revenue roughly doubled in two years, timing matching the hyperscaler capex cycle exactly.
- **ANALYSIS – HIGH** Quanta owns the scarcest input – trained craft labor at scale – and every AIDC power trend (transmission, substations, gas generation, BESS, inside-the-fence electrical) converges on that same labor pool (*Quanta dossier*).

THE INDUSTRY'S ANSWER – PREFABRICATION CONVERTS ELECTRICIANS INTO FACTORY THROUGHPUT

Table 13: Prefabrication offerings, compared on the schedule claim

VENDOR	THE PRODUCT	THE SCHEDULE CLAIM	CITATION
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VENDOR	THE PRODUCT	THE SCHEDULE CLAIM	CITATION
Rosendin	Modular Power Solutions – 7,000+ MW of annual fabrication capacity across 2M+ sq ft; a robotic solar trial running 350–400 modules per 8-hour shift with a 2-person crew, ≈3x manual	Project Goliath delivered 20 MW of electrical infrastructure in 26 weeks	<i>Rosendin dossier, 2025-04-07; sector page</i>
Quanta	Prefab and modular through Cupertino and Dynamic Systems, plus vertical integration into equipment supply – HV transformer capacity nearly doubling by 2028, the Hyosung HICO breaker JV	Schedule acceleration “in some cases by more than 12 months”	<i>2026 Investor Day, 2026-03-31</i>
Eaton	Fibrebond (\$1.45B) – engineered-to-order modular power enclosures, installed and tested off-site; plus Flexnode turnkey prefab data halls, 3.5–35 MW	≈35% schedule compression – moving competition from component price to installed-megawatt lead time	<i>release, 2025-04-01; 2026-01-28</i>
Vertiv	SmartRun; four new or expanded Americas plants taking infrastructure-solutions capacity ≈7x	Installs at ≈1 MW+ per day – up to 85% faster than stick-build	<i>2025-03-18 / 2026-03-24</i>
Huawei	PowerPOD, shipped 90%+ prefabricated	Delivery cut from 28 to 18 weeks , with 70% less onsite labor	<i>Huawei summit, 2025-05-13</i>
Delta	Prefab AI modular data center	Cuts deployment by up to 60%	<i>Computex, 2026-06-02</i>
Tesla	Megablock – four Megapack 3s plus factory-integrated MV transformer and switchgear	23% faster installation, up to 40% lower construction cost, 1 GWh deployable in ≈20 business days	<i>ESS News, 2025-09-09</i>
FlexGen	Bought commissioning firm Clean Energy Services – 125+ field technicians, 15+ GWh commissioned	70% faster installations ; buying the field-service labor that makes availability guarantees deliverable	<i>Business Wire, 2026-04-02</i>

VENDOR	THE PRODUCT	THE SCHEDULE CLAIM	CITATION
ABB	VoltaGrid synchronous condensers delivered as prefab eHouse packages	Factory-integrated stabilization	Engineering.com, 2026-03-25

ANALYSIS – MODERATE Prefab converts the binding constraint – skilled electricians – into factory throughput, and Rosendin's robotic solar trial extends the logic to field labor (Rosendin dossier). EVE makes the same argument at the cell level: its 628 Ah class claims -50% installation workload versus 314 Ah systems.

SALES TRANSLATION

Labor-per-MWh is now a spec, not a footnote. Products win when they arrive as factory-tested blocks that scarce electricians only connect – that is the actual meaning of the Megablock / HaoHan / Smartstack density race, and it is why “installed-MW lead time” (Eaton's Fibrebond logic) is replacing component price as the competitive unit. Treat the electrical contractors as both **channel and customer**: Rosendin co-developing BESSUPS with FlexGen shows the #2 contractor productizing storage itself – the natural partner, or competitor, for any BESS or UPS offering that touches the data hall.

8. THE SALES PLAYBOOK – WORKING THE 40-COMPANY MAP

8.1 THE ACCOUNT MAP – WHAT EACH OF THE 40 IS TO THIS PIPELINE

Table 14: The account map

ROLE	ACCOUNTS	THE LOAD-BEARING FACT
Spec-setter	NVIDIA	The 800 VDC partner slate <i>is</i> the qualification list for the 2027 generation; the grid tier joining it moved 800 VDC to where BESS lives ANALYSIS (<i>NVIDIA dossier</i>)
Direct power-buyer prospects	xAI, Crusoe, Oracle, OpenAI, Equinix, Google, CoreWeave	Oracle runs five simultaneously different site power architectures – each campus is a fresh competitive power decision , and its energy policy post is a de facto procurement spec ANALYSIS – HIGH . OpenAI's SB Energy tie-up means power sellers can now face OpenAI as an end-buyer, not just its landlords ANALYSIS . Equinix's choices function as procurement standards for the whole colo tier ANALYSIS – HIGH . Caveats: assume Tesla gets first look at xAI storage scope ANALYSIS – MODERATE ; CoreWeave's power decisions often sit with developer landlords – ~\$3.5B lease liabilities, and the Q3 2025 guidance cut came from one developer's delay ANALYSIS – MODERATE
Channel accounts they own the OEM decision	Constellation, Quanta, Rosendin, plus the hyperscalers' developer and utility counterparties – AES, Primergy, Entergy, DTE, Xcel, Enbridge	Amazon's utility-scale OEM decisions sit with AES and Primergy – the most established hyperscaler-linked storage channel ANALYSIS – HIGH . Meta's channel is its utility counterparties, not Meta ANALYSIS – MODERATE . Microsoft's utilities and grid partners are the channel; the Dublin grid-interactive UPS pilot with Eaton is the door still ajar for direct storage sales ANALYSIS – HIGH . Quanta and Rosendin are channel <i>and</i> customer – and, via BESSUPS and transformer self-supply, sometimes competitor

Table 15: Competitors by lane – and the dossier-documented weak point of each

LANE	WHO COMPETES THERE	KNOWN WEAK POINT TO WORK
BESS	Sungrow, Tesla, CATL, BYD, Hithium, EVE, Fluence, FlexGen, Wärtsilä (→ RCT JV), LGES, Samsung SDI, Panasonic, Jinko, Huawei	Wärtsilä's counterparty transition (≈Q3 2026 close); Fluence's execution constraint, not demand; EVE and CATL closing out of the US market; Hithium's prospectus names its embedded accounts – Jupiter Power, Lightsource bp, Samsung C&T – a map of where that competitor already sits (<i>HKEX prospectus, 2025-10-27</i>)
Rack & DC power	Delta, LITEON, Megmeet, Vertiv, Eaton, Schneider, ABB	ANALYSIS – HIGH ABB has no transformer or HVDC scope and a three-time US bribery-settlement history. ANALYSIS – MODERATE/HIGH Vertiv's binding risk is execution, and its #1 position is a statistical tie hyperscalers deliberately multi-source around. No AC incumbent has named a shipping 800 VDC switchboard

LANE	WHO COMPETES THERE	KNOWN WEAK POINT TO WORK
Prime movers	GE Vernova, Siemens Energy (→ “Omterra”), Wärtsilä engines, Bloom	GE Vernova's pricing is scarcity rent, not moat – competitor supply lands 2028–2030. Bloom's headline pipeline overstates firm demand; management disclaims visibility beyond ≈6 months
Grid	Hitachi Energy	Capacity relief does not land until 2027–28 – the gap years are the selling window ANALYSIS – MODERATE

8.2 ENTRY POINTS AND TALK TRACKS, BY THEME

- **GPU transients are a named, productized problem – open with it.** Eaton shipped market-first detection of subsynchronous oscillations *caused by AI GPU load bursting (release, 2025-09-09)*; Hitachi's CEO warns AI loads spike “up to 10 times” normal draw, and August 2026 reporting says volatile AI demand is physically damaging equipment (*Hitachi dossier, 2025-01 / 2026-08-06*). **ANALYSIS** Buffering batteries are the named remedy – a direct BESS and power-quality tailwind. Every stabilization product (grid-forming PCS, supercapacitor conditioners, flywheels, BBU) hangs off this hook.
- **Sell the schedule, not the spec.** The queue is 4–8 years; turbine slots sell for 2031 (Figure 3). Quote the compression numbers buyers already believe: Bloom 55 days, Wärtsilä 12-month plants, FlexGen under a year, Tesla Megablock 1 GWh in ≈20 business days, Huawei's 18-week PowerPOD as the international bar **ANALYSIS – HIGH** – Huawei is the excluded benchmark Western vendors get measured against.
- **Price the wedge with Quanta's math.** HV transformers and substations are 10–15% of a load-center budget, low-voltage electrical another 10–15%, at ≈\$13.5M/MW of craft-led spend (*Quanta Investor Day, 2026-03-31*) – the cleanest public numbers for sizing any power or storage scope inside a campus budget.
- **Lead with the compliance stack in the US.** FEOC and domestic content (LGES: “durable advantage is regulatory, not technical” **ANALYSIS – HIGH**), UL large-scale fire certification (Samsung SDI's deliberate premium strategy **ANALYSIS – MODERATE**), and cybersecurity attestation (Wärtsilä's IEC 62443 / SOC 2 as a procurement checkbox **ANALYSIS – MODERATE**) are now qualifying criteria, not differentiators.
- **Get specced upstream, in the twin.** Schneider's edge is the design layer – NVIDIA co-developed designs pull hardware into projects *before competitive bidding starts* **ANALYSIS – HIGH**; Fluence got Smartstack into Siemens' NVIDIA reference architecture. If the product is not in a reference design or an Omniverse DSX asset library, someone else's is.
- **Work the displacement windows.** Wärtsilä's storage counterparty changes around Q3 2026 at the RCT JV close – contract and service continuity anxiety is an opening **ANALYSIS – MODERATE**; Powin's bankruptcy shows integrator distress creates takeover paths for orphaned fleets, and FlexGen waived HybridOS fees to capture them; and every AC incumbent's unshipped 800 VDC switchboard is a challenger's proof-gap argument.

8.3 TIMING TRIGGERS – THE PIPELINE CALENDAR

Table 16: Dated checkpoints, and what each one would confirm or refute

DATE	EVENT	WHAT IT TELLS THE PIPELINE
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DATE	EVENT	WHAT IT TELLS THE PIPELINE
2026-08-11	Sinexcel H1	First AIDC revenue disclosure – whether the pre-revenue challenger tier is converting the 800 VDC reset into revenue
≈2026-08-20	EVE results	Whether pricing power held as the cell market tightened, against the three-direction US market closure
2026-08-26	NVIDIA <i>company-confirmed</i>	The demand engine's shipment cadence – the input under every third-party power-demand projection
2026-08-27	Megmeet H1	The six-fold-rebound test – whether AI-DC power revenue materialized after being called immaterial through 2025
≈2026-08-27	Jinko <i>estimate</i>	Solar-plus-storage supply into AIDC, including the 800V HVDC-compatible Sunny 365 line
2026-08-29	Sungrow + BYD	Whether the league-table reset holds – the #1 integrator and the #1 system shipper report the same day
2026-10	Samsung SDI StarPlus	US LFP cell production start – ANALYSIS – MODERATE the pivotal execution item in the compliant lane
2H 2026	Tesla Megapack 3 deliveries	Whether Brookshire's 50 GWh/yr line converts to customer volume
3Q 2026	First 800 VDC shipments	The vendor-list reset becomes measurable in revenue, not slideware
2027	Kyber racks	The qualification window closes; rack storage becomes mandatory content
2027	GE Vernova 1,000-SST decision	Whether the grid tier leapfrogs the AC-switchboard incumbents or the SST market forms around challengers
Undated	PJM co-location rulemaking	Constellation management expects data-center deal flow to “kick off with a bit of a bang” once it lands – the starting gun for the next PPA wave and its firming and storage attach

*Note: near-dated earnings dates come from the repository's armed post-earnings refresh-Routine schedule – operational scheduling data, not dossier content; the fired sessions verify actual publication. **OpenAI's announcement cadence is the sector's primary timing trigger** – Oracle, CoreWeave, AMD and Broadcom order books repriced on its announcements **ANALYSIS – HIGH**. **Housekeeping that prevents embarrassment:** Siemens Energy rebrands to “Omterra” from late 2026 – vendor names, branding and possibly tickers in proposals change within months (release, 2026-07-14).*

8.4 RISKS TO THE PIPELINE – WHAT COULD SHRINK THE MARKET

Table 17: The risk stack, to monitor on a calendar rather than a hunch

RISK	CONFIDENCE	THE EVIDENCE	THE LEADING INDICATOR
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RISK	CONFIDENCE	THE EVIDENCE	THE LEADING INDICATOR
Circular financing	High	NVIDIA↔OpenAI↔Oracle↔Amazon; Microsoft's \$4.1B quarterly equity-method losses imply ≈\$11.5B quarterly OpenAI losses	Order books – they would show a pause before the announcements do
Announced-to-operational gap	Moderate	≈0.3 GW live against more than 9 GW projected by 2029 at Stargate – delivery risk concentrated in exactly the constrained resources this report covers	Energization milestones per site, not GW announcements
Backlog duration	Moderate	Only ≈20% of the 307 GW US pipeline converts near-term; the bulk is 2028+	A capex pause shows up in order books first (<i>Eaton dossier</i>)
Political and ratepayer backlash	Moderate	Three PJM capacity auctions at the cap; a 13-governor affordability coalition; legislated clawback as a live scenario	Meta's Louisiana docket is the regulatory bellwether for how much firm generation versus storage gets built for AI load
Permitting litigation	Documented	The xAI NAACP case and Texas permit-by-rule scrutiny now shape every BTM procurement conversation	Permit status of comparable BTM gas fleets (<i>xAI, Crusoe, Oracle dossiers</i>)
Scarcity-rent normalization	Moderate	The 2027–28 capacity wave – transformers, turbines, prefab plants – can deflate pricing assumptions baked into 2026-era proposals	Table 3's commissioning dates; re-validate any quote priced for 2028+ delivery
Inverter-ban spillover	Low	If the draft FCC ban on Chinese energy inverters extends to storage and UPS power electronics, second-order effects could touch the entire Chinese AIDC supply chain serving Western markets	Scope language in the final FCC rule (<i>Huawei dossier</i>)

9. METHOD & CITATION NOTES

- **Corpus.** The 40 Profiler dossiers in the repository's profiler data set, researched 2026-08-07 to 2026-08-09 under the Source Priority Protocol – first-party investor-relations and press material exhausted before third-party – plus the Amazon revision of 2026-08-10 (profile version 2).
- **Method.** No new research was performed for this report. Eight parallel extraction agents read all 40 dossiers in full and returned 572 themed, source-tagged claims with 40-of-40 coverage verified; the report was synthesized from those claims plus direct reads of the xAI and Crusoe dossiers. Every factual claim traces to a dossier source entry or recent-development source. A four-agent adversarial verification pass then checked 353 report claims against the dossiers for figure fidelity, citation fidelity, analysis labeling and inventions; all 20 findings were corrected before publication.
- **Citations.** (*Source label, date – Company dossier*). The label matches the dossier's source entry, whose URL field holds the link; dates are source publication dates – YYYY-MM when only the month is known, absent for undated evergreen pages.
- **ANALYSIS items** come from dossier strategy-read sections, or recent-development takeaways, and are labeled inference with the dossier's own (High)/(Moderate)/(Low) confidence tags preserved. They are never presented as sourced fact, and unverified press reports retain their hedges – for example the Samsung SDI–AWS BBU talks, the Fluence-at-Bellefield inference, and the Tesla–Samsung LFP talks.
- **Field notes.** None used. The notes log lives in Drive behind Master-ACL gating and none were supplied to this session; per the schema, notes are never citable sources.
- **Ranking caveat.** BESS rankings are scope-dependent – Wood Mackenzie comprehensive versus Benchmark system shipments versus InfoLink cells versus SNE. This report names the ranking body on every ranking claim; do the same in customer-facing use.
- **Figures.** Every figure restates quantities already cited in the body text and carries its own source line. The chart palette was validated for colour-vision-deficiency separation and contrast against the page surface; ranges are drawn as hatched extensions rather than separate marks, and every donut wedge is direct-labeled so identity is never carried by colour alone.
- **Presentation styles.** This report is issued in the five writing styles registered in *PROFILER-STYLES.md* – Analyst Prose, BloombergNEF Research Report, Sell-Side Research Note, Intelligence Community Briefing, and Axios Smart Brevity – each typeset to the matching export skin in the Profiler app, so a document handed to a customer reads in the same voice as the tool it came from. **The skins change typography and section chrome only: the report text is identical across all five.** Chart colours are deliberately held constant across skins – the palette was validated once against the white print surface, and re-tinting it per style would put the data layer at the mercy of a typographic choice.
- **Freshness.** Figures are as-published through ≈2026-08-10. The armed post-earnings Routines – 27 tickers plus the private-company quarterly sweep – refresh the underlying dossiers; regenerate the affected report sections after material refreshes. Near-dated: Sinexcel H1 (08-11), EVE, NVIDIA, Megmeet, Jinko, Sungrow and BYD, all late August 2026.

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